🪂 ParaFlightLog

The Intelligent, Sovereign Paragliding Platform

A Modern Flight Analytics, Gear Lifecycle & Automation Engine Built on Open Principles

> ***"Your Flight Data. Sovereign. Intelligent. Infinitely Extensible."***
>
> *A high-performance, private, zero-cost personal flight operations center that runs entirely within your personal Google Workspace. No monthly subscriptions. No proprietary clouds. No vendor lock-in.*

1. The Manifesto: Flight Data in an Era of Walled Gardens

The Pilot's Dilemma

Every free-flight pilot understands the struggle of fragmented data. Over a flying career, your tracklogs become scattered across a dozen incompatible ecosystems:

- Dedicated flight computers (**Flymaster**, **XC Tracer**, **Oudie**, **Syride**).
- Mobile cockpit apps (**XCTrack**, **FlyMe**, **SeeYou Navigator**, **Garmin**).
- Online contest servers (**XContest**, **DHV-XC**, **Leonardo**).
- Miscellaneous folders of `.igc` and `.gpx` files lost across multiple hard drives.

When a commercial flight-tracking service introduces a monthly subscription, changes its terms, or shuts down, **your hard-earned airtime and flight history are held hostage**. Commercial flight logbooks treat your data as their proprietary asset, locking your statistics behind paywalls and denying you the ability to deeply query, customize, or extend your records.

The ParaFlightLog Vision

ParaFlightLog was born from a fundamental belief: **A pilot should have absolute sovereignty over their flight records.**

We built an enterprise-grade flight analytics, equipment maintenance, and automation engine that runs **100% inside your personal Google Drive and Google Sheets**. It combines the zero-cost scalability of cloud serverless computing with the familiar, flexible power of a spreadsheet—empowered by native **Gemini AI** to deliver insights no closed app can match.

2. Core Capabilities: Flight Logging Reimagined

PARAGLIDING FLIGHT ENGINE

📥 Ingestion	🧮 Physics	🌦️ Enrichment	📊 Cockpit
• Drag & Drop	• Thermal Detect	• Open-Meteo	• KPI Banners
• Cloud URLs	• Ridge Soaring	• Wind at Launch	• 6-Mo Currency
• Watch Folders	• Glide L/D Ratio	• Cloudbase (m)	• 8 Leaderboard
• Multi-Archive	• FAI XC Scoring	• OSM Geocoding	• Dynamic Top 5
• Smart Dedupe	• RDP Optimizer	• XContest Sync	• SeeYou 3D

📥 1. Zero-Friction Multi-Source Ingestion

- **Drag-and-Drop Web Uploader**: Drop single IGC files or compressed archives directly into the browser.
- **Continuous Drive Watch Folders**: Set up a dedicated folder in Google Drive. Drop your flight files from your phone or instrument, and ParaFlightLog automatically ingests, processes, and archives them in the background.
- **Bulk Archive Unpacking**: High-speed batch processing for `.zip` and `.gz` archives, effortlessly onboarding 500+ historical flights in minutes.
- **Smart Heuristic Deduplication**: Multi-criteria matching pairs dual-device recordings (e.g. your phone running XCTrack plus your backup audio vario), grading track quality (0–100 pts) and labeling primary vs. secondary logs without corrupting your statistics.

🧮 2. Aeronautical Physics & FAI Scoring Engine

- **Automated Flight Regime Classification**: Sliding-window physics algorithms decompose every second of your flight into **Thermal Climb**, **Dynamic Ridge Soaring**, and **Straight Glide**.
- **Climb & Glide Diagnostics**: Computes peak integrated 10s climbs, maximum sink, average thermal efficiency, and true aerodynamic glide ratios (L/D).

- **FAI Cross-Country Optimization**: Built-in Ramer-Douglas-Peucker (RDP) trajectory simplification and brute-force 3-Turnpoint optimization for **Free Distance**, **Out-and-Return**, **Flat Triangle**, and **FAI Triangle** scoring.

🌩️ 3. Meteorological & Geographic Enrichment

- **Historical Weather Backfill**: Automatically queries meteorological archives to pinpoint surface wind speeds, gusts, cardinal directions, ambient temperature, cloud cover %, and estimated boundary-layer cloudbase heights at the exact minute and coordinates of takeoff and landing.
- **Automated Site Resolution**: Uses OpenStreetMap and global paragliding site databases to automatically identify takeoff site names, landing towns, counties, and countries.

🏆 4. Competition & League Sync

- **Native XContest Probing**: Directly validates flights against XContest.org servers, links permanent competition IDs, and alerts you if a flight is approaching the strict 14-day claiming deadline.
- **One-Click 3D Visualization**: Instant hyperlink generation to inspect your flight in 3D terrain on **SeeYou Cloud**.

📊 5. Personal Cockpit & Currency Dashboard

- **Instant KPI Banners**: High-contrast career and year-to-date summaries displaying total flights, cumulative airtime hours, and cross-country kilometers.
- **Rolling 6-Month Pilot Currency Monitor**: Dynamic safety indicators (Green / Yellow / Orange) monitoring your recent airtime to ensure you stay current and safe.
- **8-Card Personal Hall of Fame**: Dynamic leaderboards celebrating your top 5 longest flights, highest altitudes, biggest thermal climbs, and peak ground speeds.

3. A Peek Under the Hood: The Genius of Google Sheets + Apps Script

Why build an advanced flight operations platform on **Google Sheets** and **Google Apps Script** instead of a traditional mobile app or web SaaS?

Traditional Flight Apps (SaaS)	The ParaFlightLog Model
Subscription Lock-in: 50–120/year forever.	**100% Free Forever**: Runs on Google's free consumer tier.
Data in Silos: Your flights reside on corporate servers.	**Absolute Sovereignty**: Stored directly in your private Google Drive.
Rigid Layouts: You can only view what the developer designed.	**Infinitely Hackable**: Create custom charts, pivot tables, and columns.
App Discontinuation Risk: If the startup fails, data is lost.	**Permanent Durability**: Standard spreadsheet and IGC files last for decades.
Limited AI: Generic stats or rudimentary summaries.	**Native Gemini AI**: Query your flight history in plain English.

🛠️ The Power of Single-Tenant Serverless

- **Zero Infrastructure Overhead**: No Docker containers to manage, no Linux updates, no database connection strings. Google manages 99.9% uptime, backup redundancy, and security.
- **Universal Multi-Device Access**: Seamlessly review and log flights on your desktop workstation, your iPad in the campervan, or your smartphone at the landing field via native Google Sheets mobile apps.
- **Complete Portability**: Your data is always ready for instant export to CSV, Microsoft Excel, GeoJSON, or JSON.

4. The AI Powerhouse: Native Gemini Intelligence

Because ParaFlightLog lives inside Google Sheets, you unlock direct integration with **Google Gemini**. You can interrogate your entire flying history using natural human language—no complex formulas required:

> 💬 *"Gemini, compare my glide ratios and ground speeds on my Ozone Rush 6 versus my previous wing when flying into headwinds exceeding 15 km/h."*

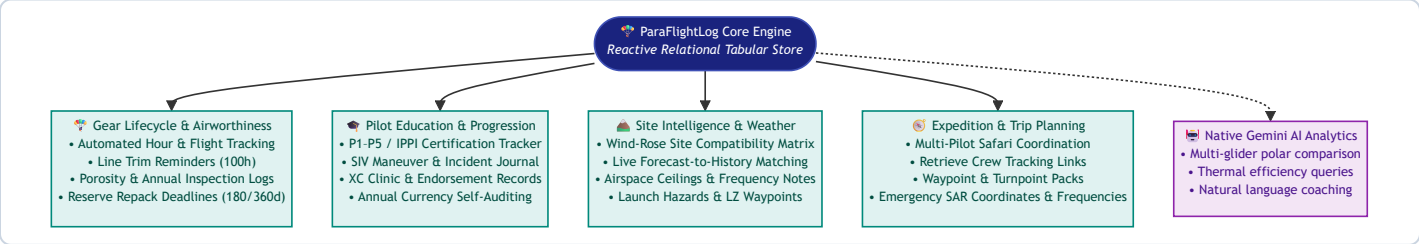
> 💬 *"Gemini, analyze all flights launched from Mount Tamalpais over the last two years. What wind direction and time of day yielded my highest thermal climb rates?"*

> 💬 *"Gemini, summarize my airtime progression, thermalling percentage, and highest altitude gains over the past 12 months in a format suitable for my FAI Advanced Pilot certification renewal."*

Gemini acts as your personal aerodynamicist and flight coach, transforming passive tracklogs into actionable flying intelligence.

5. Beyond the Logbook: An Expandable Flight Operations Platform

ParaFlightLog is engineered not merely as a historical diary, but as a **comprehensive paragliding operations center**:



🪂 1. Intelligent Equipment Lifecycle & Airworthiness

- **Automated Usage Logging**: Your gliders, harnesses, and reserves automatically accumulate flight hours and launch counts directly from your flight rows.
- **Proactive Safety Alerts**: Visual indicators flag when a wing reaches its 100-hour line trim check, when fabric porosity testing is recommended, or when a reserve parachute reaches its 180-day / 360-day repack deadline.

🎓 2. Pilot Education & Progression Matrix

- **Rating & Endorsement Tracking**: Structured competency tracking for national associations (USHPA, BHPA, DHV, HPAC) and international FAI / IPPI licensing.
- **SIV & Maneuver Journal**: Record maneuvers executed during safety clinics (asymmetric collapses, stalls, spirals) with wing-specific notes and instructor feedback.

🏔️ 3. Flying Site Intelligence & Weather Matrix

- **Centralized Site Registry**: Maintain detailed notes on launches, landing fields, frequency channels, site regulations, and airspace ceilings.
- **Predictive Weekend Matching**: Compare upcoming weekend meteorological forecasts with your personal historical flight data to recommend which local site offers the highest probability of good flying.

🧭 4. Expedition & Trip Planning

- Store launch coordinates, LZ waypoints, emergency radio frequencies, retrieve vehicle contact details, and Garmin inReach MapShare links for cross-country flying safaris.

6. Execution Freedom: The Tri-Mode Engine

Long-running flight imports and historical backfills should never lock up your computer. ParaFlightLog introduces the **Tri-Mode Execution Subsystem** (`DSGN-UI-001`):

🚀 Run Here	📌 Run in Sidebar	⚙️ Run in Background
Modal dialog execution with instant feedback. Traditional high-context dialog for quick, single-flight tasks.	**Modeless** execution docked to a compact 300px side panel. Continue browsing while progress bars update in real time.	**Headless** automated queue. Bounded by a 4.5-minute time budget, batch jobs self-schedule continuation triggers.

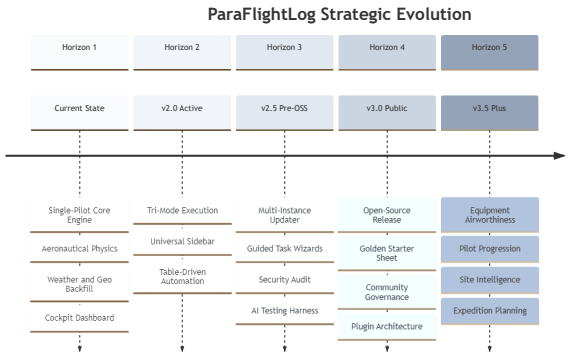
7. Where We Stand Today: Current Operational Capabilities

ParaFlightLog is currently in **active production deployment (v1.x / v2.0 Ready)**, powering daily personal flight operations with rock-solid reliability:

Capability Area	Current Operational Status	Key Subsystems & Highlights
Multi-Source Ingestion	✅ **Live & Operational**	Drag-and-drop web uploader, Drive folder/URL import, `.zip` / `.gz` extraction, continuous Watch Folder scanning, and high-speed bulk batching.
Aero Physics Engine	✅ **Live & Operational**	Sliding-window thermalling, ridge soaring, and glide detection; climb/sink rate integration; L/D aerodynamic efficiency.
FAI Cross-Country Scoring	✅ **Live & Operational**	RDP trajectory optimization, 3-turnpoint brute-force solver, Free Flight, Out-and-Return, Flat Triangle, and FAI Triangle scoring.
Meteorological Backfill	✅ **Live & Operational**	Open-Meteo historical archive API with 7-day forecast fallback; surface winds, gusts, temps, cloud cover, and Lawrence cloudbase spread.
Geographic Site Resolution	✅ **Live & Operational**	OpenStreetMap Nominatim reverse geocoding & Overpass API site detection; 6-hour caching in `CacheService`.
Contest League Probing	✅ **Live & Operational**	Direct XContest flight probing with Cloudflare bypass, UTC date shifting, permanent flight ID linking, and 14-day claim deadline countdown.
Cockpit Dashboard	✅ **Live & Operational**	Resilient 6-column layout with career/YTD KPI banners, 6-month pilot currency alerts (Green/Yellow/Orange), and 8 dynamic leaderboards.
Tri-Mode Execution	✅ **Live & Operational**	Foreground Modal (`Run Here`), Modeless Sidebar (`Run in Sidebar`), and Headless Background Queue with 4.5-minute time budgeting.
Data Integrity & Dedupe	✅ **Live & Operational**	Row-based MD5 hashing on `Flights` (Col 70), multi-criteria heuristic overlap detection, 0–100 pt sensor quality scoring, and duplicate pairing.
Configuration Management	✅ **Live & Operational**	49 parameters across 9 sections, 10 lookups, and granular 5-group JSON export/import with dynamic UI protection.

8. Strategic Roadmap: The 5 Horizons of Evolution

ParaFlightLog is evolving systematically from a private flight operations engine into a globally accessible, community-governed open-source platform:



Roadmap Highlights:

- **Horizon 2 (v2.0)**: Extends our live Tri-Mode Execution engine into a full **Table-Driven Automation Engine** (`DSGN-AUT-001`), allowing pilots to configure recurring automation triggers and pipelines from a simple spreadsheet tab.
- **Horizon 3 (v2.5)**: Introduces a **Non-Destructive Multi-Instance Updater** (`RQ-SYS-010`) to deploy code and schema updates across distributed pilot sheets without touching flight data, paired with automated AI test suites (`CR-DEV-004`) and strict PII scrubbers (`CR-SEC-001`).
- **Horizon 4 (v3.0)**: **Public Open-Source Launch** on GitHub under the MIT License, featuring a one-click **Golden Starter Sheet** that allows any pilot in the world to launch their own sovereign logbook in under 3 minutes.
- **Horizon 5 (v3.5+)**: Full expansion into **Equipment Airworthiness Management**, **Pilot Education & Progression**, and **Flying Site Intelligence**.

9. Open Source, Community & Extensibility

ParaFlightLog is **100% open source** released under the permissive **MIT License**.

- **No Walled Gardens**: Built by pilots, for pilots. The code is completely auditable, transparent, and forkable.
- **Plugin-Ready Architecture**: Easily create custom hardware parsers, connect to regional league servers, or integrate local weather stations.
- **Community-Driven Evolution**: Participate in design discussions, propose new features via RFCs, and contribute improvements back to the global free-flight community.

10. Take Flight in 3 Minutes

Getting started requires no software installation, no command line, and no server configuration:

1. **Make a Copy**: Open the public **Golden Starter Sheet** and click `File > Make a copy` into your personal Google Drive.
2. **Run the Setup Wizard**: Click `Flight Log > First-Time Setup Wizard`. The system guides you through selecting your Drive storage folders and entering your pilot name.
3. **Drop Your Flights**: Drag and drop your `.igc` files or paste a Google Drive folder link.
4. **Enjoy Sovereign Analytics**: Watch your personal dashboard populate with rich aeronautical data, weather, and leaderboards!

Fly far, fly safe, and own your sky. 🪂✨